Highlights

A signal-to-drift ratio for quantifying winding-fault detectability and selecting condition monitoring across three wind-turbine generator types

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- A drift-calibrated statistic quantifies seeded-fault detectability without a model
- Phasor residuals remove the faulted-phase dependence of negative-sequence ratios
- Signatures relocate across voltage, controller and field current by machine and drive
- A decision table selects the cheapest sensor suite meeting power 0.9 at the extent
- Detectors transfer across generators once healthy-referenced (AUC 0.85 to 0.98)